

**Step - 1 : Procedure / Implementation**

```
# Install required libraries
!pip install transformers
!pip install datasets
!pip install torch
!pip install scikit-learn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->typer)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->typer)
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->typer)
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Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
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Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0->datasets) (2025.3.0)
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Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.0)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0->datasets) (2.5.0)
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0->datasets) (25.1.0)
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer
from transformers import AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

**Step - 2 : Text Preprocessing**

```
from datasets import load_dataset

# Load dataset
dataset = load_dataset("imdb")

# Create train and test variables
train_data = dataset["train"]
test_data = dataset["test"]
```

```
# Load pretrained tokenizer (DistilBERT)
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%                               483/483 [00:00<00:00, 9.86kB/s]

tokenizer_config.json: 100%                       48.0/48.0 [00:00<00:00, 677B/s]

vocab.txt:      232k/? [00:00<00:00, 2.73MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
tokenizer.json:  466k/? [00:00<00:00, 4.35MB/s]
```

Step - 3 : Vocabulary Creation & Encoding

```
def tokenize_function(example):

    # Tokenize text
    return tokenizer(
        example["text"],
        padding="max_length",
        truncation=True,
        max_length=256
    )

# Apply tokenization to dataset
tokenized_train = train_data.map(tokenize_function, batched=True)
tokenized_test = test_data.map(tokenize_function, batched=True)
```

```
Map: 100%                               20000/20000 [00:31<00:00, 803.51 examples/s]

Map: 100%                               5000/5000 [00:05<00:00, 938.18 examples/s]
```

Step - 4 : Train-Test Split

```
dataset = load_dataset("imdb")
dataset = dataset["train"].train_test_split(test_size=0.2)
train_data = dataset["train"]
test_data = dataset["test"]
```

```
README.md:      7.81k/? [00:00<00:00, 163kB/s]

plain_text/train-00000-of-00001.parquet: 100%                21.0M/21.0M [00:01<00:00, 129MB/s]

plain_text/test-00000-of-00001.parquet: 100%                 20.5M/20.5M [00:01<00:00, 62.5MB/s]

plain_text/unsupervised-00000-of-00001.p(...): 100%          42.0M/42.0M [00:01<00:00, 43.0MB/s]

Generating train split: 100%                                25000/25000 [00:01<00:00, 24668.05 examples/s]

Generating test split: 100%                                25000/25000 [00:01<00:00, 22557.84 examples/s]

Generating unsupervised split: 100%                         50000/50000 [00:01<00:00, 45658.79 examples/s]
```

Step - 5 : Create PyTorch Dataset Format

```
tokenized_train.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"])
```

```
)

tokenized_test.set_format(
    type="torch",
    columns=["input_ids", "attention_mask", "label"]
)
```

### Step - 6 : Load Transformer Model

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2 # positive / negative
)
```

```
model.safetensors: 96%                               256M/268M [00:06<00:00, 68.3MB/s]

Loading weights: 100%                                100/100 [00:00<00:00, 321.67it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
```

| Key                     | Status     |
|-------------------------|------------|
| vocab_transform.bias    | UNEXPECTED |
| vocab_transform.weight  | UNEXPECTED |
| vocab_layer_norm.bias   | UNEXPECTED |
| vocab_layer_norm.weight | UNEXPECTED |
| vocab_projector.bias    | UNEXPECTED |
| classifier.weight       | MISSING    |
| pre_classifier.weight   | MISSING    |
| pre_classifier.bias     | MISSING    |
| classifier.bias         | MISSING    |

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

### Step - 7 : Define Evaluation Metrics

```
def compute_metrics(pred):

    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(
        labels,
        preds,
        average='binary'
    )

    acc = accuracy_score(labels, preds)

    return {
        "accuracy": acc,
        "f1": f1,
        "precision": precision,
        "recall": recall
    }
```

### step - 7 : Training Configuration

```
training_args = TrainingArguments(

    output_dir="./results",

    eval_strategy="epoch",

    learning_rate=2e-5,

    per_device_train_batch_size=8,

    per_device_eval_batch_size=8,

    num_train_epochs=2,

    weight_decay=0.01,
```

```
    logging_dir="./logs"  
)
```

`logging\_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD\_LOGGING\_DIR` instead.

### Step - 8 : Model Training

```
trainer = Trainer(  
    model=model,  
    args=training_args,  
    train_dataset=tokenized_train,  
    eval_dataset=tokenized_test,  
    compute_metrics=compute_metrics  
)
```

```
# Train the model  
trainer.train()
```

/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin\_memory' argument is set as true but  
super().\_\_init\_\_(loader)

 [ 144/5000 25:31 < 14:32:49, 0.09 it/s, Epoch 0.06/2]

| Epoch | Training Loss | Validation Loss |
|-------|---------------|-----------------|
|-------|---------------|-----------------|

### Step - 9 : Model Evaluation

```
results = trainer.evaluate()  
  
print(results)
```